Hongki Lim

Curriculum Vitae

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Professional Appointments

Inha University Mar 2022–present
Assistant Professor, Department of Electrical and Electronic Engineering

Siemens Healthineers Sep 2020–Jan 2022
Senior AI Scientist, Digital Technology & Innovation

Qualcomm Feb–Jun 2015
Intern, Computer Vision Group, Corporate Research & Development

Samsung Electronics Jul 2012–Jan 2015
Associate, Technology Planning Group, Strategic Planning Team, System LSI Division

Education

University of Michigan May 2020
Ph.D., Electrical and Computer Engineering
Concentration: Signal Processing and Machine Learning
Dissertation: *Quantitative Image Reconstruction Methods for Low Signal-to-Noise Ratio Emission Tomography*
Advisors: Jeffrey A. Fessler and Yuni K. Dewaraja

Inha University Aug 2012
B.S., Electronic Engineering

Research Interests

Probabilistic generative modeling and conditional inference; diffusion models and sampling methods for inverse problems; image reconstruction and analysis; reward-guided generation and diverse design.

Conference Papers and Proceedings

† Corresponding author.

[C10] Minwoo Kim, Seunghyeok Shin, Dabin Kim, and **Hongki Lim**†. "Tethered Predictive-Inertial Proposals with Objective Verification for Diffusion-Prior Inverse Problems." *Conference on Neural Information Processing Systems (NeurIPS)*, 2026. Accepted.

[C9] Seunghyeok Shin, Minwoo Kim, Dabin Kim, and **Hongki Lim**†. "Geometry-correct diffusion posterior sampling with denoiser-pullback curvature guidance and manifold-aligned damping." *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, PMLR 306:112591–112623, 2026.

[C8] Minwoo Kim, Seunghyeok Shin, and **Hongki Lim**†. "FAST-DIPS: Adjoint-free analytic steps and hard-constrained likelihood correction for diffusion-prior inverse problems." *International Conference on Learning Representations (ICLR)*, 2026.

[C7] Seunghyeok Shin, Dabin Kim, and **Hongki Lim**†. "Adaptive 3D reconstruction via diffusion priors and forward curvature-matching likelihood updates." *Advances in Neural Information Processing Systems (NeurIPS)*, 38, 2025. **Spotlight.**

[C6] **Hongki Lim**, Yuni K. Dewaraja, and Jeffrey A. Fessler. "Joint low-count PET/CT segmentation and reconstruction with paired variational neural networks." *Proceedings of SPIE, Medical Imaging: Physics of Medical Imaging*, 11312:113120U, 2020. Oral presentation.

[C5] Haowei Xiang, **Hongki Lim**, Jeffrey A. Fessler, and Yuni K. Dewaraja. "SPECT/CT scatter estimation using a deep convolutional neural network: Implementation in Y-90 imaging." *IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC)*, pp. 1–3, 2019.

[C4] **Hongki Lim**, Zhengyu Huang, Jeffrey A. Fessler, Yuni K. Dewaraja, and Il Yong Chun. "Application of trained Deep BCD-Net to iterative low-count PET image reconstruction." *IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC)*, pp. 1–4, 2018. Oral presentation.

[C3] Se Young Chun, **Hongki Lim**, Jeffrey A. Fessler, and Yuni K. Dewaraja. "On parameter selection for joint spectral reconstruction in Y90 SPECT." *IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC)*, pp. 1–4, 2018.

[C2] Il Yong Chun, **Hongki Lim**, Zhengyu Huang, and Jeffrey A. Fessler. "Fast and convergent iterative image recovery using trained convolutional neural networks." *56th Annual Allerton Conference on Communication, Control, and Computing*, pp. 155–159, 2018.

[C1] **Hongki Lim**, Yuni K. Dewaraja, and Jeffrey A. Fessler. "Reducing bias in Y-90 PET images by enforcing non-negativity in projection space." *IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC)*, pp. 1–4, 2017. Oral presentation.

Journal Papers

[J12] Minwoo Kim and **Hongki Lim**†. "SURE Guided Posterior Sampling: Trajectory Correction for Diffusion-Based Inverse Problems." *IEEE Access*, 14:141413–141427, 2026.

[J11] Dabin Kim and **Hongki Lim**†. "Noise-aware adaptive diffusion sampling for accelerated knee MRI reconstruction." *Physics in Medicine & Biology*, 71(10):105029, 2026.

[J10] Kyeongkyu Lee, Seonghwan Yoon, and **Hongki Lim**†. "CLARIFID: Improving radiology report generation by reinforcing clinically accurate impressions and enforcing detailed findings." *Expert Systems with Applications*, 303:130633, 2026.

[J9] **Hongki Lim**. "Direct conditional score modeling for accelerated MRI reconstruction." *IEEE Access*, 12:163914–163923, 2024.

[J8] Je-I Yang, Kwang Sub Yoon, and **Hongki Lim**†. "A 12-bit SAR ADC with binary search calibration algorithm for a split capacitor." *Electronics*, 13(2):414, 2024.

[J7] **Hongki Lim**. "Transformer-based integrated framework for joint reconstruction and segmentation in accelerated knee MRI." *Electronics*, 12(21):4434, 2023.

[J6] **Hongki Lim**†, Yuni K. Dewaraja, and Jeffrey A. Fessler. "SPECT reconstruction with a trained regularizer using CT-side information: Application to ^{177}Lu SPECT imaging." *IEEE Transactions on Computational Imaging*, 9:846–856, 2023.

[J5] Il Yong Chun, Zhengyu Huang, **Hongki Lim**, and Jeffrey A. Fessler. "Momentum-Net: Fast and convergent iterative neural network for inverse problems." *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(4):4915–4931, 2023.

[J4] Haowei Xiang, **Hongki Lim**, Jeffrey A. Fessler, and Yuni K. Dewaraja. "A deep neural network for fast and accurate scatter estimation in quantitative SPECT/CT under challenging scatter conditions." *European Journal of Nuclear Medicine and Molecular Imaging*, 47:2956–2967, 2020.

[J3] **Hongki Lim**†, Il Yong Chun, Yuni K. Dewaraja, and Jeffrey A. Fessler. "Improved low-count quantitative PET reconstruction with an iterative neural network." *IEEE Transactions on Medical Imaging*, 39(11):3512–3522, 2020.

[J2] **Hongki Lim**, Jeffrey A. Fessler, Scott J. Wilderman, Allen F. Brooks, and Yuni K. Dewaraja. "Y-90 SPECT ML image reconstruction with a new model for tissue-dependent bremsstrahlung production using CT information: A proof-of-concept study." *Physics in Medicine & Biology*, 63(11):115001, 2018.

[J1] **Hongki Lim**, Yuni K. Dewaraja, and Jeffrey A. Fessler. "A PET reconstruction formulation that enforces non-negativity in projection space for bias reduction in Y-90 imaging." *Physics in Medicine & Biology*, 63(3):035042, 2018.

Research Funding

Principal Investigator

Joint Distribution Modeling of Discrete-Continuous Mixed Variables Using Multiple Medical Datasets
National Research Foundation of Korea (NRF), RS-2026-25476632; PI: Hongki Lim
Mar 2026–Feb 2029

Diffusion Models in Gaussian Space for 3D Medical Imaging Reconstruction
National Research Foundation of Korea (NRF), RS-2025-24683103; PI: Hongki Lim
Sep 2025–Aug 2026

GPU Server for Research on Medical Image Report Generation Using Diffusion Model-based Image Synthesis
Korea Basic Science Institute (KBSI), RS-2024-00401899; PI: Hongki Lim
May 2024–Apr 2025

Unifying Image Restoration and Analysis Tasks with Iterative Neural Networks
National Research Foundation of Korea (NRF), 2022R1F1A1069055; PI: Hongki Lim
Jun 2022–Feb 2024

Participating Grants

Pioneer Program in Next-generation Artificial Intelligence for Industrial Convergence
BK21 FOUR; PI: Byung Cheol Song, Inha University
Sep 2024–Aug 2027

Developing Multimodal Generative AI Talent for Industrial Convergence
Institute of Information & Communications Technology Planning & Evaluation (IITP), RS-2024-00360227;
PI: Donghwan Kim, 42Maru
Apr 2024–Dec 2027

Graduate School of Digital-based Sustainable Energy Process Innovation Convergence
Korea Institute of Energy Technology Evaluation and Planning (KETEP), RS-2023-00243974; PI: Sungwon Hwang, Inha University
Apr 2023–Dec 2027

Artificial Intelligence Convergence Innovation Human Resources Development
Institute of Information & Communications Technology Planning & Evaluation (IITP), RS-2022-00155915;
PI: In Kyu Park, Inha University
Completed Dec 2025

Innovative Human Resource Development for Local Intellectualization Support Program
Institute of Information & Communications Technology Planning & Evaluation (IITP), RS-2023-00259678;
PI: Jaehyun Park, Inha University
Jul 2023–Dec 2024

Patents

Granted Patents

[G2] Mahmoud Mostapha, Boris Mailhe, Dorin Comaniciu, Nirmal Janardhanan, Simon Arberet, **Hongki Lim**, and Mariappan S. Nadar. “Global tractography based on machine learning.” US 12,507,959 B2, granted December 30, 2025. Assignee at grant: Siemens Healthineers AG.

[G1] Boris Mailhe, Dorin Comaniciu, Simon Arberet, Nirmal Janardhanan, Mariappan S. Nadar, **Hongki Lim**, and Mahmoud Mostapha. “Reconstruction in magnetic resonance imaging with image representations as implicit functions in time.” US 11,835,613 B2, granted December 5, 2023. Assignee at grant: Siemens Healthcare GmbH.

Patent Applications

Republic of Korea. Applicant: Inha University Industry-Academic Cooperation Foundation.

[P6] **Hongki Lim**, Seunghyeok Shin, Minwoo Kim, and Dabin Kim. “Method and Hardware Device for Signal Reconstruction Using Curvature-Aware Likelihood Correction.” Application 10-2026-0145821, filed August 5, 2026.

[P5] **Hongki Lim**, Seunghyeok Shin, and Dabin Kim. “Point Cloud Reconstruction Method and Hardware Apparatus Using Forward Curvature-Matching.” Application 10-2026-0112052, filed June 19, 2026.

[P4] **Hongki Lim**, Seonghwan Yoon, and Kyeongkyu Lee. “Method and Hardware Device for Automated Medical Image Report Generation.” Application 10-2026-0112051, filed June 19, 2026.

[P3] **Hongki Lim**, Minwoo Kim, and Seunghyeok Shin. “Signal Restoration Method and Hardware Apparatus Using Pre-trained Diffusion Model as Prior.” Application 10-2026-0112050, filed June 19, 2026.

[P2] **Hongki Lim** and Minwoo Kim. “Method for Medical Image Reconstruction.” Application 10-2025-0000786, filed January 3, 2025.

[P1] **Hongki Lim**, Minwoo Kim, and Hwan Lee. "Integrated Imaging System and Method for High-Quality Reconstruction and Region Segmentation of Undersampled MRI Data." Application 10-2024-0200661, filed December 30, 2024.

Research Datasets

[D2] **Hongki Lim** and Yuni K. Dewaraja. "Y-90 patients PET/CT & SPECT/CT and corresponding contours dataset." University of Michigan, Deep Blue Data, 2019.

[D1] **Hongki Lim** and Yuni K. Dewaraja. "Lu-177 patients CT images and contours dataset for medical image segmentation." University of Michigan, Deep Blue Data, 2019.

Conference Abstracts and Presentations

[A7] **Hongki Lim**, Il Yong Chun, Jeffrey A. Fessler, and Yuni K. Dewaraja. "Improved low count quantitative SPECT reconstruction with a trained deep learning based regularizer." *Journal of Nuclear Medicine*, 60(suppl. 1):42, 2019. Oral presentation.

[A6] **Hongki Lim**, Jeffrey A. Fessler, and Yuni K. Dewaraja. "Joint dual photopeak image reconstruction in Lu-177 SPECT." *European Journal of Nuclear Medicine and Molecular Imaging*, 45:S95–S96, 2018. Oral presentation.

[A5] **Hongki Lim**, Kyungsang Kim, Quanzheng Li, Jeffrey A. Fessler, and Yuni K. Dewaraja. "Bias reduction in Y-90 PET with reconstruction that relaxes the non-negativity constraint." *Journal of Nuclear Medicine*, 59(suppl. 1):580, 2018. Oral presentation.

[A4] Jeremy Niedbala, **Hongki Lim**, Robert Ackermann, David Hubers, Nancy A. Obuchowski, John Dickson, and Yuni K. Dewaraja. "The impact of total counts on iodine-123 ioflupane SPECT quantification." *Radiological Society of North America (RSNA) Annual Meeting*, scientific abstract SSE16-05, November 2017. Oral presentation; presenter: Jeremy Niedbala.

[A3] **Hongki Lim**, Neal Clinthorne, Maurizio Conti, Jeffrey A. Fessler, and Yuni K. Dewaraja. "Quantitative Y-90 PET for dosimetry in radioembolization." *European Journal of Nuclear Medicine and Molecular Imaging*, 44(suppl. 2):S398, 2017.

[A2] **Hongki Lim** and Yuni K. Dewaraja. "Impact of Tc-99m SPECT reconstruction methods on lung shunt and lesion/normal liver activity quantification in radioembolization." *Journal of Nuclear Medicine*, 58(suppl. 1):1032, 2017. Poster presentation.

[A1] **Hongki Lim**, Yuni K. Dewaraja, Scott J. Wilderman, and Jeffrey A. Fessler. "Y-90 SPECT maximum likelihood image reconstruction with a new model for tissue-dependent bremsstrahlung production." *Journal of Nuclear Medicine*, 58(suppl. 1):746, 2017. Oral presentation.

Research Supervision

Current Graduate Students

Minwoo Kim (Ph.D.); Seunghyeok Shin (integrated M.S.–Ph.D.); Dabin Kim, Kyeongkyu Lee, Jinhyoung Woo, Sangjun An, and Chang Gi Min (M.S.).

Completed Master's Students

Seonghwan Yoon: M.S., August 2026

Minwoo Kim: M.S., February 2026; continuing as a Ph.D. student in the lab

Hwan Lee: M.S., February 2026

Student Recognition

Seunghyeok Shin: Excellence in Research Award, BK21 FOUR, Inha University, for 2025 research; awarded February 2026.

Best Paper Award, IEEE/IEIE International Conference on Consumer Electronics-Asia (ICCE-Asia) 2025, for coauthored work presented by Hwan Lee.

Undergraduate Research Mentoring

Undergraduate Research Opportunity Program (UROP), University of Michigan, 2016–2017. Medical image reconstruction and MATLAB.

Teaching

Inha University

Undergraduate Courses

Linear Algebra: Spring 2022–2026
Introduction to Machine Learning: Fall 2022–2024; Spring 2025–2026
Deep Learning: Fall 2022–2026
Random Variables: Fall 2025–2026
Electromagnetics: Spring 2022–2024
Electrical and Electronic Engineering Capstone Design: Fall 2025–2026

Graduate Courses

Deep Generative Models: Fall 2023, 2025
Generative Artificial Intelligence: Fall 2024
AI Project III (co-taught): Spring 2025
Deep Learning: Modern Research Topics: Spring 2026
AI Convergence Project 2: Fall 2026
Machine Learning for Energy Process Innovation: Fall 2024, Fall 2026

Invited Talks

Inha University Oct 12, 2026
Invited alumni address, Graduate School Information Session, Department of Electrical and Computer Engineering

Korean Institute of Broadcast and Media Engineers Summer Conference Jun 19, 2026
Jeju, South Korea. “Adaptive 3D reconstruction via diffusion priors and forward curvature-matching likelihood updates.” Invited PI presentation

Electronics and Telecommunications Research Institute (ETRI) Feb 20, 2025
Daejeon, South Korea. Invited seminar on deep-learning-based image reconstruction

32nd Joint Conference on Signal Processing Sep 29, 2022
Ewha Womans University. “Deep-learning based image reconstruction and analysis methods for low signal-to-noise ratio emission tomography.” Invited talk

Michigan State University Feb 28, 2020
Department of Computational Mathematics, Science and Engineering. “Machine learning based image reconstruction and analysis methods for low signal-to-noise ratio emission tomography.”

Massachusetts General Hospital Dec 6, 2019
Gordon Center for Medical Imaging. “Quantitative image reconstruction and analysis methods for low signal-to-noise ratio emission tomography.”

Honors and Awards

Silver Reviewer Award, International Conference on Machine Learning (ICML) 2026
IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC) Trainee Grant 2017, 2018
Rackham Conference Travel Grants, University of Michigan 2017, 2018

Professional Service

Membership and Activities

Institute of Electronics and Information Engineers (IEIE): Executive Board Member, 2025–2026

- IEEE/IEIE International Conference on Consumer Electronics-Asia (ICCE-Asia) (2025): Financial Chair, Special Session Organizer, and Tutorial Session Chair.
- Joint Summer School on Image Understanding and Image Processing (2022): Organizing Committee Member.

Conference Reviewing

NeurIPS (2026); *ICML* (2026); *ICLR* (2027); *CVPR* (2026); *ECCV* (2026); *AAAI* (2027).

Journal Reviewing

IEEE Transactions on Medical Imaging; *IEEE Transactions on Computational Imaging*; *IEEE Transactions on Radiation and Plasma Medical Sciences*; *IEEE Journal of Biomedical and Health Informatics*; *IEEE Access*; *Pattern Recognition*; *Journal of Mathematical Imaging and Vision*; *Quantitative Imaging in Medicine and Surgery*.
Ongoing reviewer appointment: *Transactions on Machine Learning Research*, 2026.

Grant Reviewing

Grant Proposal Reviewer, National Research Foundation of Korea (NRF), 2026.

University Service

Curriculum Committee: Member, Department of Electrical and Electronic Engineering, Inha University, 2025–present

Thesis Examination Committees: Inha University; excluding own advisees

2026 6 master's theses; doctoral dissertation: Hyunjun Jo (advisor: Deok-Hwan Kim)
2025 9 master's theses; doctoral dissertation: Srinidhi Kanagachalam (advisor: Deok-Hwan Kim)
2024 9 master's theses; doctoral dissertation: S. I. M. M. Raton Mondol (advisor: Sangmin Lee)
2023 Doctoral dissertation: Shan Ullah (advisor: Deok-Hwan Kim)
2022 5 master's theses

Military Service

Republic of Korea Air Force Jul 2009–Aug 2011
Compulsory military service; Sergeant, Avionics Maintenance Battalion